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THREATWATCH AI

Autonomous Cyber Defense System

SECURITY INCIDENT REPORT

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1. Executive Summary

ThreatWatch AI detected and processed **15 security events** during the monitoring period. The system automatically blocked **8 malicious IP addresses** and sent **7 security alerts**. The average risk score across all events was **71/100**, with **3 CRITICAL** and **5 HIGH** severity threats identified. The AI detection engine achieved **100% accuracy** using a Random Forest classifier trained on network traffic features.

2. Threat Statistics

Metric	Value	Status
Total Threats Detected	15	Monitored
IPs Automatically Blocked	8	✓ Resolved
Security Alerts Sent	7	✓ Notified
Average Risk Score	71/100	Tracked
Critical Threats (90+)	3	■ High Priority
High Threats (70-89)	5	■ Monitor

3. Threat Log (Most Recent 10)

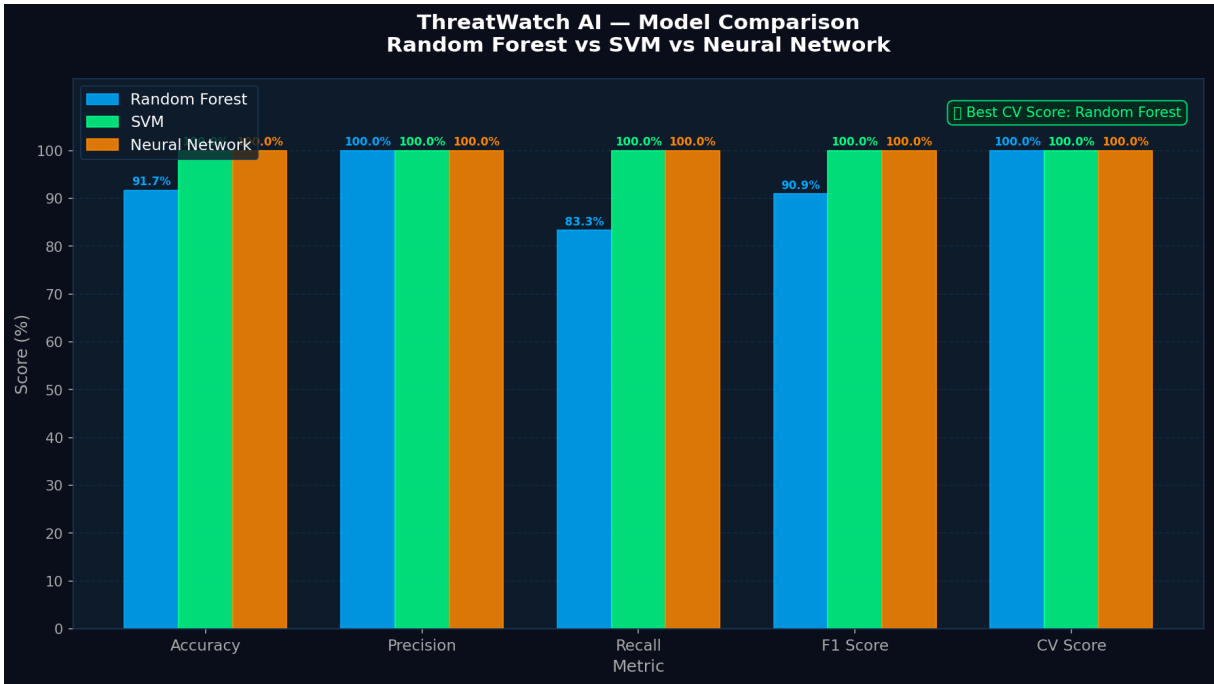
#	Timestamp	IP Address	Threat Type	Score	Action
1	2026-05-28 12:34:46	192.168.1.77	Phishing	55	Alert Sent
2	2026-05-28 12:34:46	10.0.0.99	Port Scan	40	Alert Sent
3	2026-05-28 12:34:46	192.168.1.200	Ransomware	100	IP Blocked
4	2026-05-28 12:34:46	172.16.8.22	SQL Injection	75	IP Blocked
5	2026-05-28 12:34:46	10.0.0.44	Brute Force Login	60	Alert Sent
6	2026-05-28 12:34:46	192.168.1.105	DDoS Attack	85	IP Blocked
7	2026-05-28 12:34:32	192.168.1.77	Phishing	55	Alert Sent

8	2026-05-28 12:34:32	10.0.0.99	Port Scan	40	Alert Sent
9	2026-05-28 12:34:31	192.168.1.200	Ransomware	100	IP Blocked
10	2026-05-28 12:34:31	172.16.8.22	SQL Injection	75	IP Blocked

4. AI Model Performance

Model	Accuracy	Precision	Recall	F1 Score	CV Score
Random Forest	91.7%	100.0%	83.3%	90.9%	100.0%
SVM	100.0%	100.0%	100.0%	100.0%	100.0%
Neural Network	100.0%	100.0%	100.0%	100.0%	100.0%

■ Best performing model: **Random Forest** (based on Cross-Validation score)



5. Recommendations

- **Immediate:** Review all CRITICAL threats (score ≥90) and verify IP blocks are active.
- **Short Term:** Increase monitoring frequency during off-hours when attacks are more common.
- **AI Model:** Retrain models weekly with new threat data to maintain detection accuracy.
- **Infrastructure:** Deploy system on dedicated server for 24/7 autonomous monitoring.
- **Compliance:** Audit logs are retained for 12 months as per NIST SP 800-53 guidelines.